



IEEE STUDENT BRANCH IIIT

TECHBLOCKS 12.1

ROS2 Installation Handbook

Target Setup:

Windows Subsystem for Linux (WSL) | ROS2 Jazzy

This handbook covers only the installation and verification of WSL 2, Ubuntu 24.04 LTS, and ROS 2 Jazzy.

What you will build

The TechBlox workflow uses a webcam and OpenCV to detect a colored object, convert that visual information into a simple movement command, and pass the command through ROS 2 to a simulated robot.

The workshop workflow is:

Laptop Webcam → OpenCV → HSV Color Detection → Contours → Object Position → ROS 2 Publisher → ROS 2 Topic → Subscriber → Simulated Robot / Visualization

Pre - Requisites

- A Windows 10/11 computer with administrator access.
- A stable internet connection during installation.
- Enough free disk space for Ubuntu & ROS 2.
- Permission to restart Windows when required.
- *A willingness to copy commands exactly as written.*

This handbook uses two environments:

1. PowerShell: the normal Windows terminal. Use it for WSL installation and Windows-side checks.
2. Ubuntu / WSL: the Linux terminal opened after installing WSL. Use it for ROS 2 installation and ROS commands.

WSL INSTALLATION

WSL (Windows Subsystem for Linux) allows you to run a Linux distribution such as Ubuntu directly on Windows. ROS 2 will be installed inside Ubuntu.

1	Open PowerShell as administrator. Start Menu → Search for “PowerShell” → Right Click → “Run as Administrator” On opening, you will see something <pre>Shell PS C:\Users\YourName></pre>
2	Install WSL (<i>type this inside PowerShell</i>) <pre>Shell wsl --install -d Ubuntu</pre> <i>If Windows asks you to restart, restart your computer.</i>
3	On restart, again open PowerShell and type <pre>Shell wsl</pre> Ubuntu will ask you to create a Linux username and password. Choose a username you will remember. When entering the password, <u>nothing will appear on screen while you type</u>; this is normal in Linux. To exit WSL (return back to PowerShell) <pre>Shell exit</pre>

4	Confirming the installation In PowerShell, run: <pre>Shell wsl --status wsl --list --verbose</pre> Get an output like <pre>PS C:\Users\srivt\OneDrive\Desktop> wsl --status Default Distribution: Ubuntu Default Version: 2 WSL1 is not supported with your current machine configuration. Please enable the "Windows Subsystem for Linux" optional component to use WSL1. PS C:\Users\srivt\OneDrive\Desktop> wsl --list --verbose NAME STATE VERSION * Ubuntu Stopped 2</pre>
	FROM THIS STEP WE WILL BE WORKING INSIDE WSL <u>IMPORTANT POINT</u> If your WSL opens with <pre>Shell user@user: ~\$</pre> <i>Here user@user will contain your chosen username & hostname</i> <pre>shreyansh@Shreyansh-Vivobook:~\$</pre> <i>(you are good to go, Continue with step 5)</i> ELSE If your WSL starts with anything other than what is written above For example: <pre>shreyansh@Shreyansh-Vivobook:/mnt/c/Users/srivt/OneDrive/Desktop\$</pre> YOU NEED TO CHANGE DIRECTORY TO home, Type in WSL: <pre>Shell cd ~</pre> This will change your path to: <i>(move to step5 then)</i> <pre>Shell user@user: ~\$</pre>

5	Check Ubuntu Version Shell <pre>lsb_release -a</pre> <pre>shreyansh@Shreyansh-Vivobook:~\$ lsb_release -a No LSB modules are available. Distributor ID: Ubuntu Description: Ubuntu 24.04.5 LTS Release: 24.04 Codename: noble</pre> Expected Description: Ubuntu 24.04 LTS
6	Update Ubuntu. Shell <pre>sudo apt update && sudo apt upgrade -y</pre> The sudo command runs an administrative operation. Ubuntu may ask for your Linux password. <i>(again password will be invisible when you type it)</i>
7	Install Dependencies. Shell <pre>sudo apt install software-properties-common curl git wget -y</pre>

ROS2 Jazzy INSTALLATION

ROS 2 (Robot Operating System 2) is a software framework that allows different programs and components of a robot system to communicate with one another.

1	Enable Ubuntu Universe Shell <pre>sudo add-apt-repository universe</pre>
2	Add ROS 2 repository key Shell <pre>sudo curl -sSL https://raw.githubusercontent.com/ros/rosdistro/master/ros.key \ -o /usr/share/keyrings/ros-archive-keyring.gpg</pre>
3	Add ROS 2 repository Shell <pre>echo "deb [arch=\$(dpkg --print-architecture) signed-by=/usr/share/keyrings/ros-archive-keyring.gpg] \ http://packages.ros.org/ros2/ubuntu \$(. /etc/os-release && echo \$UBUNTU_CODENAME) main" \ sudo tee /etc/apt/sources.list.d/ros2.list > /dev/null</pre>
4	Update package information. Shell <pre>sudo apt update</pre>
5	Install ROS 2 Jazzy Desktop Shell <pre>sudo apt install ros-jazzy-desktop -y</pre>

6	Configuring the ROS 2 Environment Making ROS 2 Commands available in the current terminal session. <pre>Shell source /opt/ros/jazzy/setup.bash</pre>
7	Automatically load ROS 2 <pre>Shell echo "source /opt/ros/jazzy/setup.bash" >> ~/.bashrc</pre> This adds the ROS 2 setup command to Ubuntu's shell startup file so that ROS 2 is automatically available whenever you open a new Ubuntu terminal.
8	<pre>Shell source ~/.bashrc</pre>
9	Verifying successful installation <pre>Shell ros2 pkg list head</pre> May see something like this as output: <pre>shreyansh@Shreyansh-Vivobook:~\$ ros2 pkg list head ackermann_msgs action_msgs action_tutorials_cpp action_tutorials_interfaces action_tutorials_py actionlib_msgs ament_cmake ament_cmake_auto ament_cmake_copyright ament_cmake_core</pre>

If you see this as output:

```
shreyansh@Shreyansh-Vivobook:~$ ros2 pkg list | head
ackermann_msgs
action_msgs
action_tutorials_cpp
action_tutorials_interfaces
action_tutorials_py
actionlib_msgs
ament_cmake
ament_cmake_auto
ament_cmake_copyright
ament_cmake_core
Traceback (most recent call last):
  File "/opt/ros/jazzy/bin/ros2", line 33, in <module>
    sys.exit(load_entry_point('ros2cli==0.32.12', 'console_scripts', 'ros2')())
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/opt/ros/jazzy/lib/python3.12/site-packages/ros2cli/cli.py", line 91, in main
    rc = extension.main(parser=parser, args=args)
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/opt/ros/jazzy/lib/python3.12/site-packages/ros2pkg/command/pkg.py", line 37, in main
    return extension.main(args=args)
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/opt/ros/jazzy/lib/python3.12/site-packages/ros2pkg/verb/list.py", line 24, in main
    print(pkg_name)
BrokenPipeError: [Errno 32] Broken pipe
Exception ignored in: <_io.TextIOWrapper name='<stdout>' mode='w' encoding='utf-8'>
BrokenPipeError: [Errno 32] Broken pipe
```

RELAX, YOUR ROS2 is still perfectly installed.
This traceback is not a ROS 2 installation error

CHECKING WHETHER ROS 2 ACTUALLY WORKS

Open Two separate Ubuntu Terminals

(Two separate PowerShell Terminals and type wsl in both to enter Ubuntu)

TERMINAL 1

Shell

```
ros2 run demo_nodes_cpp talker
```

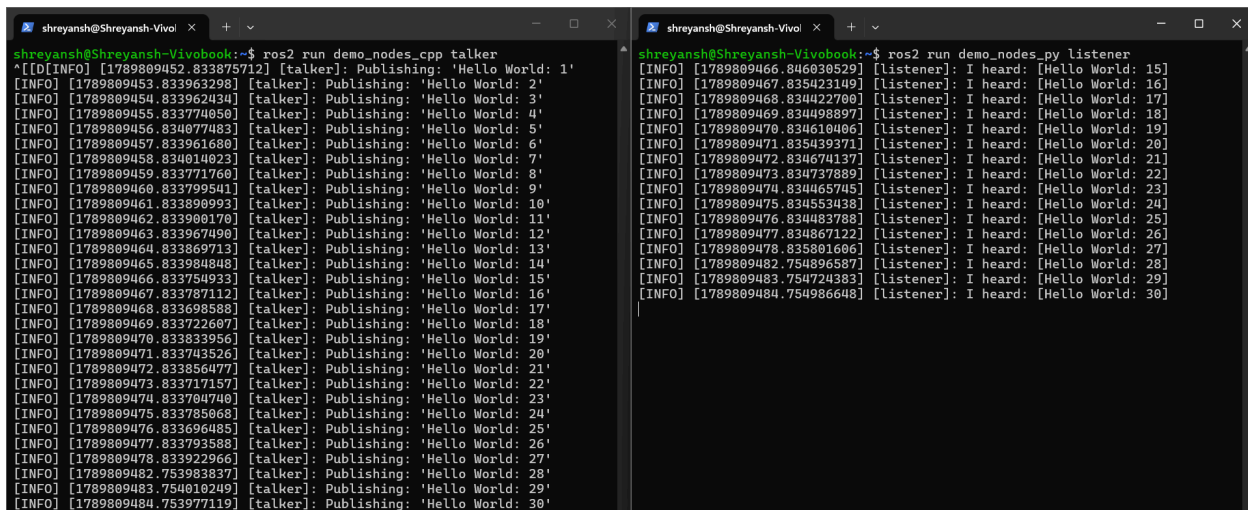
TERMINAL 2

Shell

```
ros2 run demo_nodes_py listener
```

Expected Output: Terminal 2 should continuously display received messages.

Sample Screenshot of what a successful output looks like:



The image shows two terminal windows side-by-side. The left window is titled 'shreyansh@Shreyansh-Vivobook' and shows the command 'ros2 run demo_nodes_cpp talker' being executed. It displays a series of log messages from the 'talker' node, each publishing 'Hello World' with a unique timestamp and sequence number (1 to 30). The right window is titled 'shreyansh@Shreyansh-Vivobook' and shows the command 'ros2 run demo_nodes_py listener' being executed. It displays a series of log messages from the 'listener' node, each receiving 'Hello World' with a unique timestamp and sequence number (15 to 30).

```
shreyansh@Shreyansh-Vivobook:~$ ros2 run demo_nodes_cpp talker
^[[D[INFO] [1789809452.833875712] [talker]: Publishing: 'Hello World: 1'
[INFO] [1789809453.833963298] [talker]: Publishing: 'Hello World: 2'
[INFO] [1789809454.833962434] [talker]: Publishing: 'Hello World: 3'
[INFO] [1789809455.833774850] [talker]: Publishing: 'Hello World: 4'
[INFO] [1789809456.834077483] [talker]: Publishing: 'Hello World: 5'
[INFO] [1789809457.833961680] [talker]: Publishing: 'Hello World: 6'
[INFO] [1789809458.834014023] [talker]: Publishing: 'Hello World: 7'
[INFO] [1789809459.833771760] [talker]: Publishing: 'Hello World: 8'
[INFO] [1789809460.833789541] [talker]: Publishing: 'Hello World: 9'
[INFO] [1789809461.833890993] [talker]: Publishing: 'Hello World: 10'
[INFO] [1789809462.833900170] [talker]: Publishing: 'Hello World: 11'
[INFO] [1789809463.833967490] [talker]: Publishing: 'Hello World: 12'
[INFO] [1789809464.833869713] [talker]: Publishing: 'Hello World: 13'
[INFO] [1789809465.833984848] [talker]: Publishing: 'Hello World: 14'
[INFO] [1789809466.833754933] [talker]: Publishing: 'Hello World: 15'
[INFO] [1789809467.833787112] [talker]: Publishing: 'Hello World: 16'
[INFO] [1789809468.833698588] [talker]: Publishing: 'Hello World: 17'
[INFO] [1789809469.833722607] [talker]: Publishing: 'Hello World: 18'
[INFO] [1789809470.833833956] [talker]: Publishing: 'Hello World: 19'
[INFO] [1789809471.833743526] [talker]: Publishing: 'Hello World: 20'
[INFO] [1789809472.833856477] [talker]: Publishing: 'Hello World: 21'
[INFO] [1789809473.833717157] [talker]: Publishing: 'Hello World: 22'
[INFO] [1789809474.833704740] [talker]: Publishing: 'Hello World: 23'
[INFO] [1789809475.833785068] [talker]: Publishing: 'Hello World: 24'
[INFO] [1789809476.833696485] [talker]: Publishing: 'Hello World: 25'
[INFO] [1789809477.833793588] [talker]: Publishing: 'Hello World: 26'
[INFO] [1789809478.833922966] [talker]: Publishing: 'Hello World: 27'
[INFO] [1789809482.753983837] [talker]: Publishing: 'Hello World: 28'
[INFO] [1789809483.754010249] [talker]: Publishing: 'Hello World: 29'
[INFO] [1789809484.753977119] [talker]: Publishing: 'Hello World: 30'

shreyansh@Shreyansh-Vivobook:~$ ros2 run demo_nodes_py listener
[INFO] [1789809466.846030529] [listener]: I heard: [Hello World: 15]
[INFO] [1789809467.835423149] [listener]: I heard: [Hello World: 16]
[INFO] [1789809468.834422700] [listener]: I heard: [Hello World: 17]
[INFO] [1789809469.834498897] [listener]: I heard: [Hello World: 18]
[INFO] [1789809470.834610406] [listener]: I heard: [Hello World: 19]
[INFO] [1789809471.835439371] [listener]: I heard: [Hello World: 20]
[INFO] [1789809472.834674137] [listener]: I heard: [Hello World: 21]
[INFO] [1789809473.834737889] [listener]: I heard: [Hello World: 22]
[INFO] [1789809474.834465745] [listener]: I heard: [Hello World: 23]
[INFO] [1789809475.834553438] [listener]: I heard: [Hello World: 24]
[INFO] [1789809476.834483788] [listener]: I heard: [Hello World: 25]
[INFO] [1789809477.834867122] [listener]: I heard: [Hello World: 26]
[INFO] [1789809478.835801606] [listener]: I heard: [Hello World: 27]
[INFO] [1789809482.754896587] [listener]: I heard: [Hello World: 28]
[INFO] [1789809483.754724383] [listener]: I heard: [Hello World: 29]
[INFO] [1789809484.754986648] [listener]: I heard: [Hello World: 30]
```

To stop both processes, press **Ctrl + C** in both the terminals.

yayy!!

ROS2 successfully installed :)

End of Document